CO1-AT2

THUVIKSHA.V
1924260035
Artificial Intelligence
CEA1701

1. Intelligent agent will observe the environment through sensors and acts on it. In a self ~~ob~~ smart home, sensors might detect motion, temperature, light and thermostats and door locks carry out actions

(i) Simple Reflex Agent:-

It acts only on the current input using condition-action rules. For example: if motion is detected after sunset, turn on the hallway light. It is fast and simple. But does not remember earlier events or adapt well to changing circumstances

(ii) Model-Based Agent:-

Keeps an internal model of the home and uses it to interpret current sensor readings. It could remember that someone is home even if motion sensors are temporarily inactive or track that a room is warming up. This helps when information is incomplete, though the model must be kept accurate

Scanned with OKEN Scanner

Goal based agent

choose action that help achieve a stated goal, such as keeping the living room with a comfortable temperature range or ensuring the doors locked at night. It can consider sequences of actions but may not know which of several goal-achieving option is best

Utility Based Agent

Scores possible outcomes according to factors such as comfort, energy use and security, then selects an action with the highest overall value. For example: It might a slightly warmer Thermostat setting when the home is empty to save energy while prioritizing comfort when occupants return.

Hybrid Approach

It would work best simple reflex rules can handle urgent events as smoke detection or an open door at night. A model-based component can track occupancy and environmental changes, while a utility based component balances comfort, energy efficiency and security. safety rules should remain fixed for critical situations

robot can represent each location as a state available moves as actions and exist is the goal. Because the maze is unknown it must explore, record what observes and uses that information to guide later decisions. The search algorith assume the robot can identify its current location and remember discovered paths

Uniform cost search

UCS explored the path with the lowest total cost so far. It is useful when moves have different costs.

Advantage: Finds a least-cost path when stops are non-negative

Disadvantage: can use substantial time and memory because it stores many frontier paths

Iterative Deepening Search

IDS repeatedly turforms depth-limited, depth-first searches, increasing the deep limit each time until it finds the exit.

Advantage: Uses little momory and is complete when the branching factor is finite

disadvantage: Repeats exploration near the start at each new depth limit. It is not suited to finding at least cost path when have different costs

A simple reflex agent might choose an move based only on what is currently senses. In an unfamiliar maze, that can Lead to repeated moves or loops because it does not remember explored locations. A model based, based agent can maintain a map of known corridors, walls and visited locations. In either case, the agent must update its map when it discover a new part of the maze.